

厦门大学本科课程教学大纲

课程名称	微积分 I-2		所属课程组	微积分 I	
课程代码	140080030002		英文类别代号	MATH 1027.06	
适用专业	本科理工类各专业				
课程类型	公共基本课		课程课型	理论课	
总学分	总学时	理论教学学时	实验教学学时	实践教学学时	
6	90	90	0	0	
先修课程	无				
一、课程简介					
微积分 I-2 是理工类本科各专业学生的一门必修的重要基础课,是课程微积分 I-1 的延续。 主要内容包括:微分方程,空间解析几何,级数,多元函数微积分学的基本概念、基本理论和基本运算技能,为学习后继课程奠定必要的数学基础。					
二、培养目标					
修完本课程后,学生应能掌握多元微积分的基本概念、基本理论和基本运算,特别是微分和积分的基本思想和各类积分的计算,可求解基本常微分方程,判别无穷级数敛散性以及用函数项级数表示函数。本课程旨在帮助学生发展抽象思维、逻辑推理和数学分析的技能,促使其注重数学方法和具备一定的计算能力,可以在一定程度上利用所学知识来分析和解决今后的实际问题。 此外,本课程还将涉及微积分的一些科普知识,包括涉及到的数学家事迹、理论知识的当前一些应用和研究热点,让学生学习过程中能感受到数学之美和数学家孜孜不倦的探索精神。					
三、课程思政目标					
在微积分教学和学习中,引导学生体会事物间的相互转化和对立统一的辩证关系,培养学生用辩证唯物主义的观点科学地看待问题和解决问题,提高学生逻辑思维的能力和分析能力,让学生感知到数学之美,树立正确的人生价值观和勇于探索的科学精神,落实国家育人的战略举措。					

四、教学方法
在教学过程中，采用结合实际应用例子和几何直观引入概念和数学结果，使学生更容易理解和接受关键点。除此之外，我们将准备课堂练习，随堂小测和讨论，引导学生巩固和深入知识点，并要求其课后能独立完成作业。微积分是一门系统的学科，通过一系列衔接的课堂授课以及适量的科普知识，可以激发学生的兴趣，掌握微积分最基本的一些思想。 进一步地，我们还结合多媒体设备进行教学，有利于每位老师的课堂教学规划和实现教学高效率，同时学生通过看一些动画或者视频资料，促进他们对知识点的理解和掌握，拓展他们的视野。

五、主要内容及学时安排			
章（或节）	主要内容	课程思政映射与融入点	学时安排
第七章	微分方程的基本概念，可分离变量的微分方程，齐次方程，一阶线性微分方程，可降阶的高阶微分方程，高阶线性微分方程，常系数齐次线性微分方程，常系数非齐次线性微分方程。	培养学生利用常微分方程解决实际问题的能力	14
第八章	向量及运算（线性运算、数量积、向量积、混合积、向量的坐标表示与运算、方向数与方向余弦）；空间曲面方程；平面方程和直线方程。	拓展空间想象能力，掌握解析几何的一些基础知识和运算	10
第九章	多元函数的概念、几何意义；二元函数的偏导数和全微分的概念及计算，以及有界闭区域上连续函数的性质；方向导数与梯度的概念并掌握其计算方法；多元函数极值和条件极值的概念。全微分形式的不变性，隐函数（包括由方程组确定的隐函数）的偏导数；曲线的切线和法平面及曲面的切平面和法线的方程。	诱导学生认识一元微分学和多元微分学的辩证统一关系、延拓思路 and 需要克服的困难，培养学生探索精神	18
第十章	二重积分、三重积分的概念和性质，二重积分（直角坐标、极坐标）、三重积分（直角坐标（投影法、截面法）、柱面坐标、球面坐标）的计算方法。	通过讲解概念和计算方法，使学生意识到积分原始思想的重要性和如何在处理问题过程中能化繁为简。	12

第十一章	两类曲线积分、两类曲面积分的概念和性质；计算两类曲线积分、两类曲面积分的方法；格林公式并会运用平面曲线积分与路径无关的条件；高斯公式、斯托克斯公式；应用高斯公式计算曲面积分。可适当讲述有关场论基本公式的一些科普知识。	通过讲述格林、高斯和斯托克斯等数学家故事，鼓励学生敢于创新和探索。同时引导学生认识各类积分的关系以及在流体力学中的重要性。	18
第十二章	常数项级数收敛、发散以及收敛级数的概念；级数的基本性质及收敛的判别方法；幂级数的收敛半径、收敛区间及收敛域的求法；求幂级数在收敛区间内的和函数；将一些简单函数间接展开成幂级数。	通过级数展开讲述不同函数（如多项式、三角函数）的逼近效果，进而介绍一些高精尖技术的发展，如航空航天技术的发展，以及中国科学家冯康的故事。	12
复习	期中、期末复习		6
合计			90
	六、考核方式与要求		
	统一时间，统一试卷，笔试，闭卷。最终成绩按平时成绩 50%（包含期中成绩、考勤和作业）和期末成绩 50%的比例进行计算。		
七、选用教材/教辅材料		教材，《高等数学》第七版下册，高等教育出版社，ISBN 978-7-04-039662-1，2014-8-1，“十二五”普通高等教育本科国家级规划教材，同济大学数学系。	
八、主要文献资料或相关数据库			
九、课程网站等支持条件			
十、其它信息			
	大纲制定者：	大纲审定者：	大纲制定时间：2025 年 12 月 22 日

XMU Undergraduate Course Syllabus

Course name	Calculus I-2		Course group	Calculus	
Course code	140080030002		Category code	MATH 1027.06	
Course can be shared with	Undergraduate course for science and engineering				
Course type	Basic Common Courses		Course focus	Lecture	
Credit	Total learning hours	Lecture	Experiment		Practical
5	90	90	0		0
Prerequisites	Calculus I-1				
1.Course description					
As a continuation of Calculus I-1, Calculus I - 2 is a compulsory and important basic course for science-technology students. Main contents include: differential equation, spatial analytic geometry, series, calculus of multi-variable function, and these contents' basic concepts, basic theory and basic computing skills, lay a necessary mathematical foundation for learning subsequent courses.					
2. Learning goals (including moral education goals)					
After finishing this course, students should be able to master the basic concept, elementary theory and fundamental operations of one variable calculus, especially the basic ideas of differential and integral and the calculation of various integrals. They can solve basic ordinary differential equations, judge the convergence or divergence of infinite series and express functions by functional series. This course aims to help students develop skills in abstract thinking, logical reasoning and mathematical analysis. Students are also supposed to develop skilled calculation skills and capabilities to analyze and solve real problems. In addition, this course will also involve some popular science knowledge of Calculus, including mathematician anecdotes, current applications and research hotspots of theoretical knowledge, so that students can feel the beauty of mathematics and the tireless exploration spirit of mathematicians.					
3. Moral education goals					
In the teaching and learning of calculus, we will guide students to experience the dialectical relationship of mutual transformation and unity of opposites between things, cultivate students to treat and solve problems scientifically with the viewpoint of dialectical materialism, improve students' ability of logical thinking and analysis, enable students to perceive the beauty of mathematics, establish correct values of life and the scientific spirit of courage to explore, and implement the strategic measures of national education.					

4. Teaching approaches

In the teaching process, concepts and mathematical results are introduced by combining practical application examples and geometric intuition to make it easier for students to get the key points. In addition, we will prepare classroom exercises, quiz and discussion in class, guide students to consolidate and deepen their knowledge points, and require them to complete their homework independently after class. Calculus is a systematic subject. Through a series of connected classroom teaching and an appropriate amount of popular science knowledge, students can stimulate their interest and master some of the most basic ideas of calculus. Furthermore, we also combine multimedia equipment for teaching, which is conducive to each teacher's classroom teaching planning and teaching efficiency. At the same time, students can promote their understanding and mastery of knowledge points and expand their horizons by watching some animation or video materials.

5. Content outline of the course (including moral education elements)

Chapter(Section)	Content	Main point of moral education	Learning hours
Chapter 7	concept of differential equation; variable separable equations; homogeneous equation; first order linear equation; higher order differential equation which can be reduced; solution's structure of the second order linear differential equations; second order constant coefficient homogeneous linear differential equation and its general solution; second order constant coefficient non-homogeneous linear differential equation and its solution.	Cultivate students' ability to solve practical problems by utilizing ODE	14
Chapter 8	The concept and operations of vector (linear operations, dot products, cross products, mixed product, vector coordinate representation, direction vector and direction cosine); the concepts of surfaces in space; equations of lines and planes.	Expand the ability of spatial imagination and master some basic knowledge and operations of analytic geometry	10

Chapter 9	The concept of multivariate functions and the geometric meaning; the concepts of partial derivative and total differential of bivariate functions and their calculation, and the properties of continuous functions on closed-region; the concepts of direction derivative and gravity and master their calculating method; the concept of extremum and conditional extremum of multivariate functions; understand and the total differential form invariance; partial derivatives of implicit function (implicit functions determined by equations); tangent line's and normal plane's equations of curve, tangent plane's and normal line's equations of surface.	To enable students to understand the dialectical unity relationship, extension ideas and difficulties to be overcome between single variable and multi variable differential calculus, and cultivate students' exploration spirit	18
Chapter 10	The concepts and properties of double integral and triple integral, the calculating methods of double integral (rectangular coordinate, polar coordinate), triple integral (rectangular coordinate(projection method, section method), cylindrical coordinate, spherical coordinate).	By explaining the concepts and calculation methods, students will be aware of the importance of the original idea of integral and how to reduce complexity to simplicity in the process of solving problems.	12
Chapter 11	The concepts of two types of curve integral and two types of surface integral; the calculating methods of two types of curve integrals and two types of surface integrals; Green's formula and the condition of curve integrals independent of path; Gauss's formula, Stokes' formula; calculate surface integrals using Gauss's formula.	Encourage students to innovate and explore by telling stories of mathematicians such as Green, Gauss and Stokes. At the same time, guide students to understand the relationship of various integrals and their importance in fluid mechanics.	18

Chapter 12	The concepts of convergence and divergence of infinite series, sum of infinite series; the basic properties of convergent series and necessary condition for convergent series; the calculating methods of radius, interval and domain of convergence for power series; find sum function of some power series on convergent interval; power series expansions for some simple functions.	The approximation effects of different functions (such as polynomial and trigonometric functions) are described through series expansion. At the same time, the development status of some advanced and sophisticated technologies, such as the development of aerospace technology, and the story of Chinese scientist Feng Kang are introduced.	12
General review			6
Total			90
6. Assessment methods and requirements			
Unified time, unified examination, closed book, written examination			
7.Textbooks/ References	Textbook, Advanced mathematics, Volume II, 7th Edition, Higher Education Press, ISBN 978-7-04-039662-1, 8/1/2014, National Planning Textbook for Undergraduate Higher Education in the 12th Five-Year Plan, Department of Mathematics, Tongji University.		
8.Major literature or related databases			
9.Website			
10.Others			
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